



FERMI ESTIMATION · PROJECT DOCUMENTATION

# GUESSTIMATE COACH

I guess... ✗ → AI guess... ✓

Economic logic, design decisions, challenges and limitations  
Structured as Situation · Task · Action · Result

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BUILT ON: GUESSTIMATION · GUESSTIMATION 2.0 · HACKING THE CASE INTERVIEW

## SITUATION

# Practicing estimation has no feedback loop

Guesstimates (Fermi problems) are a standard screening and reasoning tool in consulting, product management, and quantitative interviews, and a everyday skill for real decisions made under incomplete information — sizing a market, staffing a launch, sanity-checking a budget line. The standard way people practice today is static: solving problems out of a book with a fixed answer key, or rehearsing with a mentor whose time is scarce and expensive. Neither gives a learner real-time coaching through the reasoning process, and neither checks the final number against a current, real-world figure.

## TASK

# Build a self-serve, grounded coaching loop

Build a lightweight tool that (1) teaches the structured-estimation method already codified across three respected sources — *Guesstimation*, *Guesstimation 2.0*, and *Hacking the Case Interview's* 5-step framework — (2) coaches a learner through each step of their own reasoning without doing the reasoning for them, and (3) grounds the final guess against a live, real-world reference figure instead of a static answer key.

## ACTION

# What was built, and the problems solved along the way

The app walks a user through five coached steps — Clarify, Structure, Estimate, Calculate, Sanity-Check — then a sixth Verify stage. Each step's coaching prompt is sourced directly from the underlying books: order-of-magnitude rounding and anchoring on known reference quantities from the two *Guesstimation* books, and the MECE structuring / top-down vs. bottom-up decision from *Hacking the Case Interview*.

## Challenges faced, and how each was tackled

- **Keeping the AI coach from doing the thinking:** an open-ended assistant tends to just answer the question. Tackled by constraining the coaching system prompt to short, step-scoped nudges that explicitly forbid numeric estimates or full answers, and that react to the learner's own draft rather than issuing generic advice.
- **No clean API exists for a single ground-truth number:** there is no service that returns “the correct answer” to an open-ended estimation question. Tackled by pulling live Google Custom Search snippets and having the model synthesize a reference estimate from that unstructured text, returned as strict JSON so the app can parse it reliably.
- **Grading approximate answers fairly:** guesstimates are judged on order of magnitude, not precision, so exact-match grading would be meaningless. Tackled by encoding an explicit Fermi tolerance rule directly into the grading prompt — a guess within roughly 2–3x of the inferred reference counts as a sound estimate.
- **Distinctive visual design inside a templated component library:** Streamlit ships with a fixed widget set and a generic default look. Tackled with injected CSS against stable component identifiers, a custom

ruler-tick progress tracker and ink-stamp verdict badges built from raw HTML, and an original hand-drawn SVG watermark (a compass over a tick-marked ruler) in place of any external artwork.

- **Keeping the tool zero-infrastructure and privacy-respecting:** avoiding a backend or stored user data. Tackled by requiring the user's own Anthropic and Google API keys, entered at runtime and held only in session memory for that run.

## RESULT

# Economic rationale, honestly bounded

### Why it matters — the economic logic

Fast, order-of-magnitude reasoning is a cheap substitute for judgment under uncertainty: most real decisions have to be made before precise data exists, so the ability to get within 2–3x of the right answer quickly has outsized value relative to its cost of acquisition. Practicing this skill has traditionally required scarce, expensive resources — case-interview coaching commonly runs well into double-digit hourly rates and is bottlenecked by mentor availability. Replacing that bottleneck with an AI coaching loop drops the marginal cost of a single practice session to a small fraction of that, removing the scarcity constraint on the one thing that actually builds the skill: repeated, structured practice with feedback. Grounding each answer against live search results rather than a fixed textbook key also keeps the feedback current — populations, prices, and market sizes drift over time, and a static answer key silently goes stale while a live one does not.

### Limitations

- Requires the user's own paid Anthropic and Google Custom Search API keys — not a zero-cost tool, and cost scales (modestly) with usage.
- Verification quality is bounded by search result quality: for obscure or ambiguous questions the snippets may be thin, returning a low-confidence or UNKNOWN reference figure.
- Grading checks only the final numeric guess, not the validity of the intermediate reasoning — two very different reasoning paths can both land a GOOD verdict.
- No persistence across sessions by design (privacy and simplicity) — no history, streaks, or long-term progress tracking.
- Single-user, local-run tool: no authentication, multi-tenancy, or hosting layer included.
- The “within 2–3x is GOOD” threshold is a reasonable heuristic drawn from Fermi-problem convention, not a rigorously derived statistical standard.

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